

10 Ans: D

$$pV = NkT \rightarrow N = (1.00 \times 10^5)(2.20 \times 10^{-3})/kT_0$$

$$\text{microscopic KE} = 3/2 kT$$

$$\text{Total internal energy} = 3/2 N(kT_{50}) = 3/2 (1.00 \times 10^5)(2.20 \times 10^{-3})(T_{50}/T_0) = 390 \text{ J}$$

11 Ans: C